ASSOCIATED GRADED MODULES OF TWO APPROXIMATING COMPLEXES AGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The Artin-Rees perturbation set-up with the constant c working for both maps g and g'. the author cites Lemma lemma-approximate-complex (tag 07VE) in the statement, source lines 194-254. Supplies the defining data and the fact that the same constant works for both maps.
- Associated graded module of a filtered module. used at proof line 263 to express the graded pieces as subquotients of the target. Supplies the objects that are being compared.
- The hypothesis that g and g' agree modulo the higher power of the ideal. used at proof lines 279-281 in the form g = g' modulo I^{c+1}N. Is what makes the perturbation of the element land in the higher filtered piece.

Source reference. Original lemma 07VF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10005–10005. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07VE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10004–10004. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Assumptions as in Lemma 07VE. Let* $Q = \mathrm{Coker}(g)$ *and* $Q' = \mathrm{Coker}(g')$. *Then* $Gr_I(Q) \cong Gr_I(Q')$ *as graded* $Gr_I(A)$*-modules.*

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Proof. In degree n we have $\mathrm{Gr}_I(Q)_n = I^nN/(I^{n+1}N + g(M) \cap I^nN)$ and similarly for Q'. We claim that

$$g(M) \cap I^nN \subset I^{n+1}N + g'(M) \cap I^nN.$$

By symmetry (the proof of the claim will only use that c works for g which also holds for g' by the lemma) this will imply that

$$I^{n+1}N + g(M) \cap I^nN = I^{n+1}N + g'(M) \cap I^nN$$

whence $\mathrm{Gr}_I(Q)_n$ and $\mathrm{Gr}_I(Q')_n$ agree as subquotients of N, implying the lemma. Observe that the claim is clear for $n \leq c$ as $g = g' \bmod I^{c+1}N$. If $n > c$, then suppose $b \in g(M) \cap I^nN$. Write $b = g(a)$ for $a \in I^{n-c}M$. Set $b' = g'(a)$. We have $b - b' = (g - g')(a) \in I^{n+1}N$ as desired. $\square$